Algorithm 1 SPORK with Learned First-Launch Timing

Input: History H, tools $\mathscr{T}$, confidence θ, attempts R, retry step s; controller C_ϕ, decision prefix b, wait budget Δ

$M \leftarrow \text{AsyncMain}(H, \mathscr{T})$; spec $\leftarrow \varnothing$
Wait for b main tokens, or a main action/end
if a b-token reasoning prefix is available **then**
  $h \leftarrow M_{\le b}$; $(Q_{\text{now}}, Q_{\text{wait}}) \leftarrow C_\phi(H, \mathscr{T}, h; b, \Delta)$
  $Q_{\text{off}} \leftarrow 0$; $u \leftarrow \arg\max_{a \in \{\text{off,now,wait}\}} Q_a$
  if $u = \text{wait}$ **then** $h \leftarrow \text{WaitPrefix}(b + \Delta)$
  for $r = 0, \ldots, R-1$ **while** $u \neq \text{off}$ **do**
    if $r > 0$ **then** $h \leftarrow \text{WaitPrefix}(n + s)$
    if main action started or stream ended **then break**
    $n \leftarrow |M|$; $(p, c) \leftarrow \text{Fork}(H, \mathscr{T}, h)$ *D1: shared prefix KV*
    if $\text{Valid}(p)$ **and** $\text{Active}(M)$ **then** $\text{RegisterDraft}(M, p)$ *D3*
    if $\text{Valid}(p)$ **and** $c \ge \theta$ **then**
      spec $\leftarrow (p, \text{AsyncIsolatedTool}(p))$; **break** *D2*
    end if
  end for
end if
$o \leftarrow \text{AwaitValidOutcome}(M)$
if o is a tool call **then**
  $z \leftarrow \text{Commit}(\text{Await}(\text{spec}))$ **if** $\text{ExactMatch}(\text{spec}, o)$ **else** $\text{Tool}(o)$
else
  $z \leftarrow o$ *final answer*
end if
Drain pending work; clear D3 state; **return** z

Blue: one timing decision; C_ϕ predicts time saved relative to off. The main stream continues in parallel; $(b, \Delta) = (32, 64)$.

$\text{WaitPrefix}(k)$ waits for k main tokens or a main action/end, then returns the visible prefix. Fork returns a candidate call p and name-span confidence $c = \min_j e^{\ell_j}$. D3 drafts are target-verified; ExactMatch checks the tool name and arguments. Cleanup runs on every exit, including failure or cancellation.